

Assignment: Problem Statement · Lab 2

1 Scenario

You are building a small internal tool for the customer support team of an online retail platform. Every day the team receives dozens of support tickets through the help center: shipping complaints, refund requests, product defect reports, and general questions, all landing in one shared queue. Right now, a human agent has to open every ticket, figure out what it's actually about, and write a reply from scratch, which is slow and the tone/quality varies a lot from agent to agent.

Your job is to build an AI-powered support ticket triage assistant using LangChain that automatically reads an incoming ticket, works out what kind of issue it is, and drafts an appropriate first response, the same overall pattern you practiced with the reference files, applied to a new problem.

2 Your Task

Build a Python application using LangChain that:

- Classifies an incoming support ticket into exactly one of four categories:
 - **shipping_issue** : the order is delayed, lost, stuck in transit, or delivered to the wrong address
 - **refund_request** : the customer wants a refund, return, or exchange for an order
 - **product_defect** : the item arrived damaged, faulty, or not working as intended
 - **general_inquiry** : a general question about the product, order status, or policy that isn't a complaint
- Routes the message to a different prompt/template per category and generates a contextual reply using the LLM.
- Prints the input message, the classification result, and the generated reply clearly to the terminal.

3 Category-Specific Behaviour (required)

Each category's prompt must make the model behave differently, this is the actual point of the assignment, so make each response meaningfully distinct instead of a generic "thanks for reaching out":

Category	What the reply should do
shipping_issue	Acknowledge the delay/loss, apologize for the inconvenience, and ask for the order number and shipping address if not already given.
refund_request	Acknowledge the request, ask for the order number and reason for the return if missing, and explain that it will be reviewed against the refund policy.
product_defect	Apologize for the faulty item, ask for a photo/description of the defect and the order number, and reassure them a replacement or refund will be arranged.
general_inquiry	Answer helpfully and directly if possible, or point them to where they can find the answer (FAQ, order tracking page, etc.).

4 Deliverables

- A single Python file implementing the full pipeline (classification + routing + reply generation).
- Four screenshots, one run per category, each showing input, classification result, and generated reply.